

Rahul Anil Kumar

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Professional Summary

Python AI Developer specializing in Agentic AI and multi-agent systems using LangGraph, LangChain, and CrewAI. Experienced in building dynamic, tool-augmented workflows with Human-in-the-Loop validation, typed state management, and autonomous reasoning. Proficient in developing Retrieval-Augmented Generation (RAG) pipelines, and delivering scalable AI systems integrated with REST APIs, RabbitMQ, MySQL, and other microservices. Adept at translating complex LLM-powered logic into production-ready architectures.

Education

B.Tech – Artificial Intelligence and Data Science

Oct 2020 – May 2024

Rajagiri School of Engineering and Technology, Kakkanad
CGPA: 6.67

Experience

Software Developer – Teknowmics Knowledge Soln

Apr 2025 – Present

- Built RAG pipelines using LangChain and vector search tools.
- Developed LangGraph workflows and CrewAI agents for multi-agent task execution and validation.
- Created MCP-based server-client-host infrastructure for modular and scalable AI execution.
- Migrated search infrastructure from legacy Solr to Solr 9 with schema validation.
- Maintained microservices including: REST APIs, RabbitMQ queues, MySQL integrations.
- Independently developed and maintained a document Preview Service.
- Performed DevOps tasks: process control, log tracking, and deployment debugging.

Data Science Intern – Luminar Technolab, Kochi

Jun 2024 – Mar 2025

- Built foundational data pipelines in Python using pandas/numpy.
- Applied ML algorithms to real-world data for prediction and analysis.

ML Intern – NeST Digital, Aluva

Oct 2022 – Nov 2022

- Worked with TensorFlow and Scikit-learn to build ML models and interpret results.
- Focused on feature engineering and model performance enhancement.

Projects

LangGraph AI Agent Framework (2025)

- Developed LangGraph multi-agent flow with CrewAI for reasoning, validation, and tool use.
- Built a Leave Application Assistant with Human-in-the-Loop approval flows.
- Created dynamic Pydantic model generator for typed state control and validation.
- Integrated semantic memory and action-based validation workflows.

Solr 9 Migration Automation (2025)

- Migrated Solr indexes and cores from legacy systems to Solr v9 with schema compatibility.

SUPPLYSYNC – Demand Forecasting Platform (2023–24)

- Built LSTM-based time series models (82% accuracy) and KNN clustering for segmentation.

Technical Skills

- **Languages:** Python, C++, C
- **AI/ML:** LangChain, LangGraph, CrewAI, RAG, TensorFlow, Scikit-learn, NLP
- **Microservices:** REST APIs, RabbitMQ, Preview Service, MySQL
- **Data & Viz:** Pandas, Numpy, Matplotlib, Seaborn, PowerBI

Languages

- English
- Hindi
- Malayalam